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Arijit De

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Work Experience

Machine Learning Engineer | **mVizn Pte. Ltd.** | **Jan 2024 – Present**
Smart Property Manager Project | Singapore (Remote)

- Engineered a **production-grade GenAI multi-agent platform** for the construction domain, enabling Autonomous Project Management & Scheduling (APMS), Automated Estimating & Procurement (AEP), and intelligent project information retrieval.
- Designed and orchestrated **AI agent workflows** using CrewAI, enabling task delegation, reasoning, autonomous decision-making, project monitoring, and automated report generation.
- Integrated enterprise systems through **Model Context Protocol (MCP)**, tool-calling frameworks, and external APIs for real-time workflow automation.
- Built scalable **Retrieval-Augmented Generation (RAG)** pipelines leveraging vector embeddings, hybrid search and re-rankers with **Pinecone** for contextual retrieval across large volumes of construction documents, contracts, schedules, and project records.
- Applied **LoRA-based parameter-efficient fine-tuning** techniques to adapt Large Language Models (LLMs) for domain-specific understanding of CAD drawings, BIM files, and technical documentation.

3D Scan-to-BIM Project

- Led the development of an **end-to-end machine learning pipeline** for 3D semantic segmentation of point cloud data using Python and PyTorch.
- Implemented **MLflow-based experiment tracking, model versioning, model registry, and reproducible training workflows** to support scalable MLOps practices.
- Developed automated data processing and workflow orchestration pipelines using **Apache Airflow**, improving reproducibility and deployment efficiency.
- Designed and implemented backend services using **FastAPI**, Pydantic, REST APIs, and scalable software architecture principles.
- Deployed AI services using **Docker and Google Cloud Platform (GCP)**, including Cloud Storage, Compute Engine VMs, IAM and Cloud SQL services.

Machine Learning Engineer | **Synerjix** | **Jun 2025 – Dec 2025**
Smart Moving Assistant | USA (Remote)

- Created a **production-grade Smart Moving Assistant** that helps users looking to move houses list out and estimate the cost of shifting of all movable items using photos of the rooms taken using any device.
- Used **DinoV3** for object detection, **python** and **FastAPI** for backend development, **MySQL** as the Database and **AWS** (Sagemaker, EC2, EKS) as the deployment platform along with **CI/CD** and **Docker**.
- Automation improved moving estimation by 80% and helped stakeholders improve business efficiency by around 50%.

Machine Learning Engineer | **Mercedes-Benz Research and Development India** | **Aug 2018 – Aug 2019**
Advanced driver-assistance system (ADAS) | Bengaluru, India

- Developed and optimized a computer vision pipeline for Vulnerable Road User (VRU) detection using **YOLOv3** and large-scale annotated datasets.
- Designed model training, evaluation, and validation workflows to support deployment-ready perception systems for Level-3 ADAS applications.
- Built automated quality assurance and evaluation frameworks using Python, improving annotation throughput by 20% and data quality by 25%.

Systems Engineer | **Tata Consultancy Services** | **Aug 2014 – Aug 2015**
Ultimatix Team | Kolkata, India

- Maintained and enhanced high-availability backend systems (Java, MySQL) for TCS's internal website, ensuring seamless functionality for over 100,000 users.
- Streamlined feature release cycle, reducing deployment time by 20%

Skills

- Languages: Python, C++, SQL
- AI/Agent Systems: Multi-Agent Systems, Agent Orchestration (Crew AI, Google ADK), Tool Calling/Execution, Shared State/Memory, Production AI Systems, LangChain, LLM APIs (Gemini, Claude)
- Cloud & Ops: Google Cloud Platform (GCP), Amazon Web Services (AWS), Git, Docker
- Other: Scalable System Design, System Architecture, Workflow Orchestration

Projects

- **Doctor Appointment Scheduler** (Mar '26 - Apr '26) -Architected and deployed a scalable, production-level Multi-Agent System for appointment scheduling. Designed the Agent Orchestration and multi agent delegation and coordination using Google ADK, integrating intelligent doctor matching and calendar scheduling (task hand-off/execution). Implemented robust Tool Calling and Execution using Composio for calendar actions and Google Maps MCP. The scalable Backend Engineering architecture utilizes Google Cloud Run for containerized deployment, integrating Google Cloud SQL and Gemini Enterprise.
  GitHub - https://github.com/arijitde92/Doctor_Appointment_Scheduler
  Live Link - https://doctor-appointment-app-173427564927.asia-south1.run.app/

- **Job Application Agent** (Apr '25 - May '25) - Developed a real-world Multi-Agent System for automation, utilizing Crew AI for advanced workflow orchestration of specialized agents. Engineered Shared State/Memory by integrating Google BigQuery (vector store) and Vertex AI (embeddings) for consistent, scalable semantic search and storage across agents. Implemented custom tools (LinkedIn/GitHub scrapers) and leveraged LangChain for core system logic, all built in Python.
  GitHub - https://github.com/arijitde92/Job_Application_Agent

Education

- PhD, Jadavpur University, Kolkata, India. 2019–2024
- M.Tech. Computer Science & Engineering, Jadavpur University, Kolkata, India. GPA - 8.83 2016–2018
- B.Tech. Computer Science & Engineering, Techno India, Kolkata, India. GPA - 8.81 2010–2014

Publications

1. **A. De**, A. Santra, M. Tiwari and A. S. Chowdhury, "Predicting Genetic Markers for Brain Tumors Using a Composite Loss," in *IEEE Transactions on Computational Biology and Bioinformatics* (**TCBB**), 2025 July. doi: 10.1109/TCBBIO.2025.3593318.
2. **A. De**, A. S. Chowdhury, (2025). Shape Induced Multi-class Deep Graph Cut for Hippocampus Subfield Segmentation. In: Antonacopoulos, A., Chaudhuri, S., Chellappa, R., Liu, CL., Bhattacharya, S., Pal, U. (eds) Pattern Recognition. (***ICPR 2024)***. Lecture Notes in Computer Science, vol 15313. Springer, Cham. https://doi.org/10.1007/978-3-031-78201-5_16
3. **A. De**, M. Tiwari, & A. S. Chowdhury. 3D Hippocampus Segmentation Using a Hog Based Loss Function with Majority Pooling. In 2023 IEEE International Conference on Image Processing (***ICIP***), Kuala Lumpur, Malaysia ***2023***, October, pp. 2260-2264.
4. **A. De**, R. Mhatre, M. Tiwari and A. S. Chowdhury, "Brain Tumor Classification from Radiology and Histopathology using Deep Features and Graph Convolutional Network," 2022 26th International Conference on Pattern Recognition (***ICPR 2022***), Montreal, QC, Canada, *2022*, pp. 4420-4426
5. **A. De**, M. Tiwari, E. Grisan, A.S. Chowdhury, "A Deep Graph Cut Model for 3D Brain Tumor Segmentation", 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (***EMBC 2022***).
6. **A. De**, A.S. Chowdhury, "DTI based Alzheimer's Disease Classification with Rank Modulated Fusion of CNNs and Random Forest", ***Expert Syst. Appl***. 169 (*2021*), 114338. DOI: 10.1016/j.eswa.2020.114338

Certifications

- Deep Learning, a 5-course specialization, by Deeplearning.ai. Verify here.
- MCPS: Microsoft Certified Professional, Microsoft Certification No.: 1042C4-5DHF30